TiDAR vs AR Throughput (Updated Single-Forward)

Data files loaded dynamically:

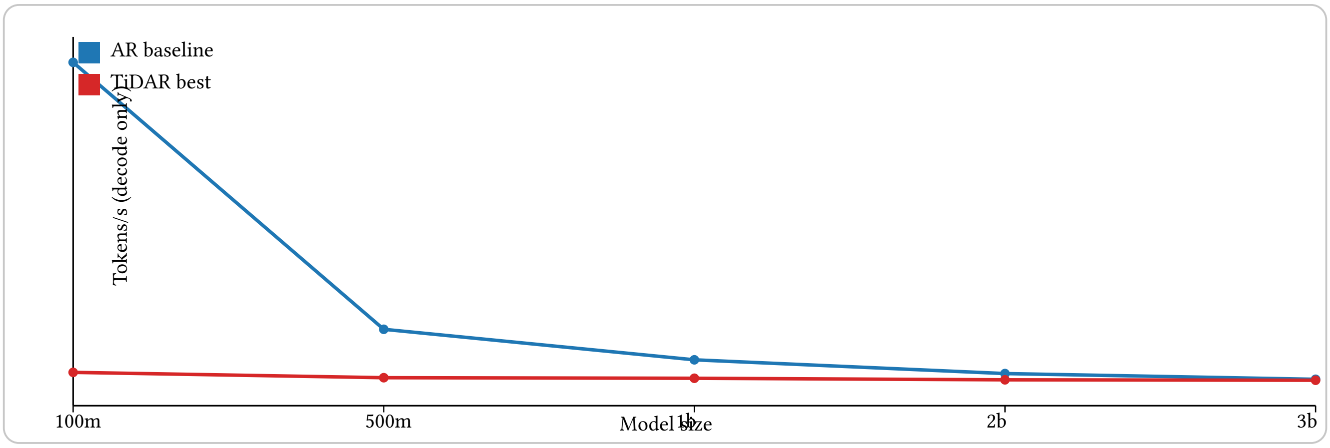
- Benchmark_logs/tidar_vs_ar_single_forward_20260208_134219 + "/best_tidar_vs_ar_by_model_prefill.json"
- Benchmark_logs/tidar_vs_ar_single_forward_20260208_134219 + "/notes.txt"

Sweep:

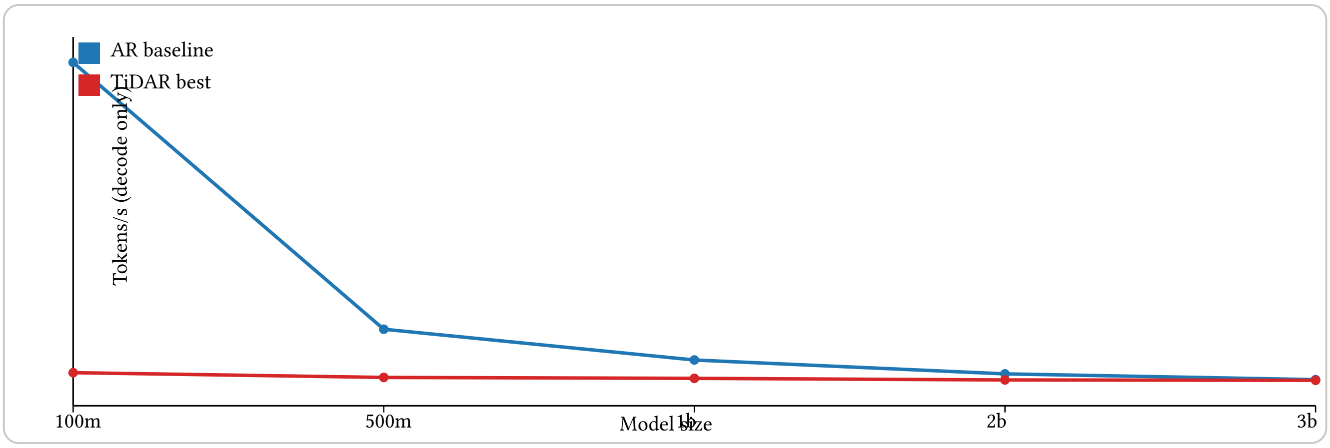
- Models: 100m, 500m, 1b, 2b, 3b
- TiDAR grid: draft_len {4, 8, 16}, accept target {0.6, 0.8}, prefill {0, 1024, 4096}
- Decode steps: 256, repeat: 2
- AR baseline: reused prior run

Line graphs: tokens/s vs model size

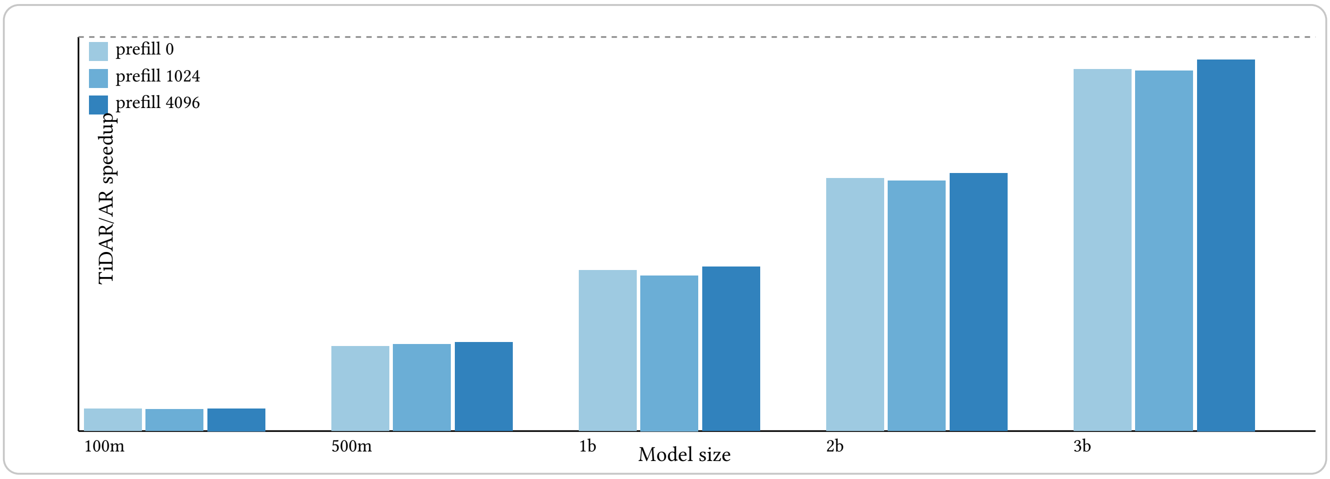
Prefill = 0



Prefill = 4096



Histogram-style grouped bars: TiDAR/AR speedup



Key findings

- Best pair: model 3b, prefill 4096, speedup 0.943x, TiDAR draft 16 at accept 0.8.
- Worst pair: model 100m, prefill 1024, speedup 0.056x.
- Trend in this setup: speedup rises with model size and higher draft, but stays below 1.0x on A40 in current JAX path.

Provenance

Updated TiDAR single-forward sweep vs AR baseline
created_at: 2026-02-08T13:42:19.639340
tidar_summary: TiDAR/Docs/Benchmark_logs/tidar_updated_narrow_20260208_094343/summary.csv
ar_summary: TiDAR/Docs/Benchmark_logs/ar_simple_baseline_sweep_20260207_103142/summary.json
best_global_speedup: 0.9431 at model=3b prefill=4096 draft=16 accept=0.8
worst_global_speedup: 0.0558 at model=100m prefill=1024 draft=4 accept=0.6
pairs_compared: 15